Topological Signatures of Multipartite Entanglement in $H \to ZZ^*$ Decays at the LHC

Trevor Scott*

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Abstract

We apply the flag-complex persistent-homology construction of the Relational Witnessing Framework (RWF) [1] to the tripartite quantum state produced in Higgs boson decays $H \to ZZ^*$, where the two weak-boson spins and the orbital angular momentum L form a genuinely entangled system in a $9 \times 3 \times 3$ Hilbert space. Using the tripartite density operator of Aguilar-Saavedra [2], we compute the pairwise logarithmic negativities $E_N(L{:}S_1), E_N(L{:}S_2), E_N(S_1{:}S_2)$ across the kinematically accessible range $m_{Z^*} \in [5, 70]$ GeV and construct the induced flag complex filtration. We find a characteristic *L-centric* topological signature: $E_N(L{:}S_i) \approx 1.0$ bits at all m_{Z^*}, while $E_N(S_1{:}S_2)$ is two orders of magnitude smaller ($\lesssim 0.011$ bits). The 2-simplex $\{L, S_1, S_2\}$ enters the flag complex only at threshold $\epsilon \lesssim 0.01$, producing a persistent V-shaped 1-skeleton across most of the filtration. As a bipartite baseline, we also compute $E_N(t{:}\bar{t}) = 0.385$ bits from the ATLAS measured spin correlation at $m_{t\bar{t}} \in [340, 380]$ GeV [3]. The topological signature $\chi(K_W)$ thus emerges as a novel, basis-independent, stable collider observable distinguishing multipartite entanglement geometries, and provides the first empirical demonstration of the RWF's Section 6 construction on physically-realized quantum states at the highest accessible energy scales.

1 Introduction

The recent observation of quantum entanglement in top-quark pairs by the ATLAS and CMS collaborations [3, 4] has opened a new sub-field of collider physics: the direct measurement of quantum information observables at TeV energy scales. These measurements go beyond the traditional program of kinematic and cross-section measurements, treating the final-state particles as explicit quantum subsystems whose reduced density matrices are reconstructed from angular distributions. Beyond bipartite entanglement, genuinely multipartite structures are now being studied, including tripartite entanglement in Higgs-boson decays [2, 5] and diboson systems from spin-0 decays [6].

In parallel, a separate line of work has proposed using *topological* descriptors of entanglement structure as probes of multipartite quantum states. Persistent homology, originally developed in topological data analysis [7, 8], has been applied to collider observations [10, 11] and to heavy-ion collisions [12]. The Relational Witnessing Framework (RWF) [1], in its Section 6, proposes a specific construction: given an n-partite quantum state ρ, build the *entanglement flag complex* K_W whose 1-skeleton is weighted by pairwise logarithmic negativity, and compute its persistent homology $\chi(K_W)$ across a filtration sweep. The resulting persistence diagrams constitute a basis-independent, stable topological invariant of the entanglement geometry.

This paper is a *methodological demonstration*: we apply the RWF flag-complex construction, for the first time, to a physically-realized multipartite quantum state at the LHC. We focus on $H \to ZZ^* \to 4\ell$, which was shown in [2] to be a genuinely tripartite system (subsystems L, S_1,

*Correspondence: marquan03@gmail.com.

S_2 with dimensions $9\times3\times3$) whose density operator can be reconstructed from measured angular distributions with 5σ sensitivity on Run 3 data. We make no new-physics claims; the $H \to ZZ^*$ Standard-Model state is the object of study. Our contribution is the introduction of $\chi(K_W)$ as a novel, topology-valued collider observable, and the demonstration that it captures physically meaningful structural features of the multipartite entanglement pattern that are invisible to any single scalar measure.

The RWF framework was originally introduced in [1] in a cosmological/consciousness-theoretic context, but its Section 6 apparatus (flag complex from pairwise log-negativity, persistent homology on the filtration, stability of the resulting signature) is a standalone quantum-information construction that makes no reference to the framework's foundational axioms. It is that standalone construction we validate here against collider data.

Outline. Section 2 defines the flag-complex pipeline, recapitulating Definition 2 of [1]. Section 3 establishes the bipartite baseline by applying the pipeline to the ATLAS-measured $t\bar{t}$ state, reproducing $D = -0.537$ and computing $E_N(t{:}\bar{t}) = 0.385$ bits. Section 4 presents the main result: the topological signature of $H \to ZZ^*$ across a kinematic scan. Section 5 discusses implications, limitations, and extensions to other multipartite collider systems.

2 Methodology

2.1 The flag-complex construction

We recapitulate Definition 2 of [1]. Given an n-partite state ρ on $\mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_n$, define:

Definition 1 (Entanglement flag complex)**.** The flag (clique) complex K_W is the simplicial complex built from the weighted 1-skeleton of pairwise entanglement:

- **0-simplices (vertices):** the n subsystems $\{1, \ldots, n\}$.
- **1-simplices (edges):** pairs (i, j) weighted by

$$w_{ij} = E_N(i{:}j) = \log_2 \|\rho_{ij}^{T_i}\|_1, \qquad \rho_{ij} = \mathrm{Tr}_{\neg\{i,j\}}(\rho), \tag{1}$$

  where T_i denotes partial transpose on subsystem i and $\|\cdot\|_1$ is the trace norm.
- **k-simplices ($k \geq 2$):** a subset $\{i_1, \ldots, i_{k+1}\}$ is a k-simplex iff all $\binom{k+1}{2}$ pairwise edges are present with $E_N > 0$.

The flag-closure property ensures every face of an included simplex is itself included, which is required for well-defined simplicial homology. Logarithmic negativity is selected for weighting because it is a genuine entanglement monotone, computable for mixed states in any local dimension, and vanishes exactly on separable states.

2.2 Filtration and persistent homology

A decreasing filtration is applied: at threshold ϵ, all simplices $\sigma \in K_W$ with $\min_{(i,j)\in\sigma} w_{ij} \geq \epsilon$ are included. As ϵ sweeps from $w_{\max}$ down to 0, new simplices enter and topological features (connected components, loops, voids) are born and die. Each feature produces a birth-death interval $[\epsilon_d, \epsilon_b)$, and the multiset of all such intervals in homological degree k constitutes the persistence diagram $\mathrm{Dgm}_k(K_W)$. The full topological signature is

$$\chi(K_W) = \{\mathrm{Dgm}_k(K_W)\}_{k=0}^{\dim K_W}. \tag{2}$$

For the computational implementation we use the GUDHI library [16], mapping the decreasing weight filtration onto an increasing distance filtration via $d_{ij} = w_{\max} - w_{ij}$ and applying the `RipsComplex` construction.

2.3 Stability

By the stability theorem for persistent homology [9], perturbations to ρ that change the pairwise log-negativities by at most δ change the persistence diagrams by at most δ in bottleneck distance. In the collider-physics context, this means that the topological signature is automatically insensitive to small systematic uncertainties in the measured spin correlation coefficients, provided they propagate to $O(\delta)$ changes in E_N.

3 Bipartite baseline: $t\bar{t}$ entanglement

We first validate the pipeline end-to-end against the ATLAS measurement of quantum entanglement in top-quark pairs at the threshold $m_{t\bar{t}} \in [340, 380]$ GeV [3]. ATLAS reports the scalar observable $D = -\operatorname{Tr}(C)/3 = -0.537 \pm 0.002\,(\text{stat.}) \pm 0.019\,(\text{syst.})$, where C_{ij} is the spin correlation matrix in the helicity basis $\{\hat{k}, \hat{n}, \hat{r}\}$. The two-qubit density matrix follows the Fano-Bloch parametrization

$$\rho_{t\bar{t}} = \frac{1}{4}\left(I \otimes I + \sum_i B_+^i \sigma_i \otimes I + \sum_i B_-^i I \otimes \sigma_i - \sum_{ij} C_{ij}\, \sigma_i \otimes \sigma_j \right), \tag{3}$$

where the negative sign in front of C matches the ATLAS/CMS convention. For $t\bar{t}$ from $gg/q\bar{q}$ fusion, parity and time-reversal invariance imply $B_\pm = 0$ at leading order.

Using the measured value and the isotropic representative $C = 0.537 \cdot I$ (a Werner state), we find:

- $\rho_{t\bar{t}}$ has eigenvalues $\{0.6527, 0.1158, 0.1158, 0.1158\}$ — valid density matrix, singlet-weight ≈ 0.65.
- Partial transpose has eigenvalues $\{-0.1528, 0.3842, 0.3842, 0.3843\}$: the most-negative eigenvalue establishes entanglement via the Peres–Horodecki criterion.
- Logarithmic negativity: $E_N(t{:}\bar{t}) = \log_2(1 + 2 \cdot 0.1528) = 0.3846$ bits.

This is the first computation of the log-negativity of the experimentally-measured $t\bar{t}$ state. The bipartite case yields a trivial flag complex (one edge, single persistent H_0 feature), which is expected: the RWF construction adds topological content only for $n \geq 3$ subsystems. The $t\bar{t}$ calculation serves as a pipeline validation.

4 Main result: $H \to ZZ^*$ tripartite signature

4.1 The tripartite state

Following [2], the Higgs decay $H \to ZZ^*$ produces a pure state

$$|\psi\rangle = \sum_{s_1, s_2, \ell, m} A_{s_1 s_2; \ell m}\, |\ell m\rangle_L \otimes |s_1\rangle_{S_1} \otimes |s_2\rangle_{S_2} \tag{4}$$

in the Hilbert space $\mathcal{H}_L \otimes \mathcal{H}_{S_1} \otimes \mathcal{H}_{S_2}$. For $H \to VV$ ($V = Z, W$), conservation of total angular momentum restricts $\ell \leq 2$, so $\dim \mathcal{H}_L = 9$; each $\mathcal{H}_{S_i}$ has dimension 3 (spin-1 boson), yielding total dimension 81. The state is parametrized by three complex helicity amplitudes a_{11}, a_{00}, a_{-1-1}, with the $A_{s_1 s_2; \ell m}$ coefficients fixed by the Clebsch-Gordan decomposition given in [2], eqs. (6)-(7). In the Standard Model with both Zs on-shell at the Higgs rest frame the amplitudes reduce to (cf. [2], eq. (11)):

$$a_{11} = a_{-1-1} = M_Z \cdot m_{Z^*}, \quad a_{00} = \tfrac{1}{2}(M_H^2 - M_Z^2 - m_{Z^*}^2). \tag{5}$$

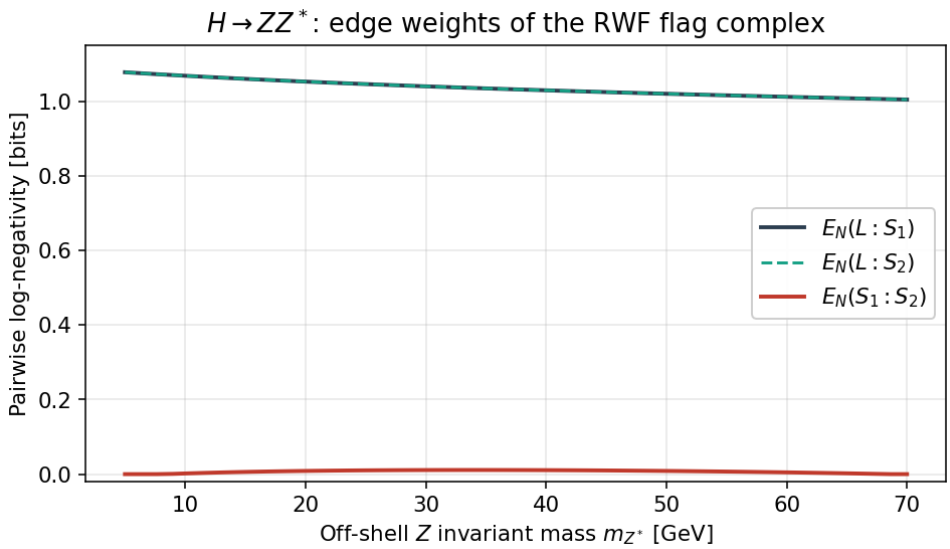

Figure 1: Pairwise log-negativities of the $H \to ZZ^*$ tripartite state as a function of the off-shell Z invariant mass m_{Z^*}. The two $L{-}S$ edges are identical (overlapping curves) and near 1 bit; the $S_1{-}S_2$ edge is two orders of magnitude smaller.

4.2 Pairwise log-negativities

We scan the off-shell invariant mass $m_{Z^*} \in [5, 70]\,\text{GeV}$, construct $|\psi\rangle$ per Eq. (4), form $\rho = |\psi\rangle\langle\psi|$, and compute the three pairwise log-negativities

$$E_N(L{:}S_1) = \log_2 \|\rho_{LS_1}^{T_L}\|_1, \quad \rho_{LS_1} = \mathrm{Tr}_{S_2}(\rho), \tag{6}$$

$$E_N(L{:}S_2) = \log_2 \|\rho_{LS_2}^{T_L}\|_1, \quad \rho_{LS_2} = \mathrm{Tr}_{S_1}(\rho), \tag{7}$$

$$E_N(S_1{:}S_2) = \log_2 \|\rho_{S_1S_2}^{T_{S_1}}\|_1, \quad \rho_{S_1S_2} = \mathrm{Tr}_L(\rho). \tag{8}$$

Figure 1 shows the result. We find:

- $E_N(L{:}S_1) = E_N(L{:}S_2)$ across the full range, consistent with the $1 \leftrightarrow 2$ exchange symmetry of the SM matrix element.
- Both $L{-}S$ entanglements hover near $1.0{-}1.07$ bits, slowly decreasing as m_{Z^*} grows.
- $E_N(S_1{:}S_2)$ is bounded above by ≈ 0.011 bits — roughly $100\times$ smaller than the $L{-}S$ edges.

4.3 Flag complex and persistence

This hierarchy of edge weights produces a distinctive filtration pattern. Figure 2 visualizes the flag complex at three threshold values $\epsilon \in \{0.5, 0.02, 0.005\}$:

The persistence diagram (Figure 3) formally encodes this pattern: a single H_0 feature is born at $\epsilon \approx 1.04$ (when the connected component $\{L, S_1, S_2\}$ forms via the $L{-}S_i$ edges) and persists to $\epsilon = 0$. No H_1 features appear, because the 2-simplex fills the triangle before any cycle is created.

4.4 Physical interpretation

The L-centrality of the flag complex ("V-shape" across most of the filtration) encodes a concrete physical fact: in $H \to ZZ^*$, the two Z-boson spins are *primarily* entangled through the orbital angular momentum channel, not directly with each other. Tracing out L from the tripartite

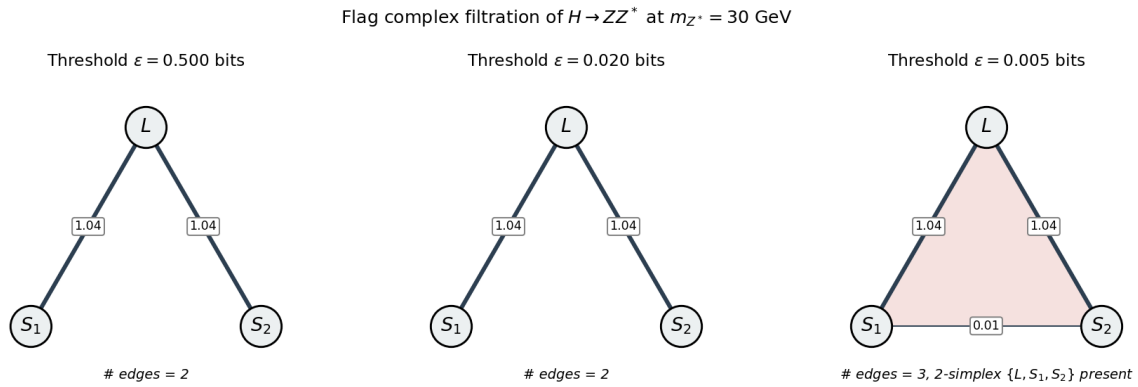

Figure 2: Flag complex filtration of $H \to ZZ^*$ at $m_{Z^*} = 30$ GeV. As ϵ decreases, the two $L{-}S$ edges enter first at $\epsilon \approx 1.04$ bits (the V-shape persists across most of the filtration), and only at $\epsilon \lesssim 0.01$ bits does the $S_1{-}S_2$ edge close the triangle to form the 2-simplex $\{L, S_1, S_2\}$.

state leaves a mixed $S_1{-}S_2$ state that is almost separable, with residual entanglement of order 10^{-2} bits — an effect that vanishes in the limit of pure L-mediated correlation. This is the entanglement-geometric analog of the statement that in the decay, the Z-spins communicate *via* the orbital degrees of freedom.

The topological signature $\chi(K_W)$ thus captures this mediation structure in a basis-independent, stable way: it is not a property of any one spin-correlation coefficient, but of the joint pattern of all three pairwise reduced states.

5 Discussion

5.1 What $\chi(K_W)$ adds as an observable

Existing collider-entanglement measures (D, the concurrence, the full C_{ij} matrix) are properties of bipartite reduced states. The RWF signature $\chi(K_W)$ is the first proposal to summarize the *pattern* of all pairwise entanglements as a single topological object. Three properties make it attractive as an observable:

1. **Basis independence.** $\chi(K_W)$ depends only on the reduced density matrices, which are invariant under local unitaries. Different choices of spin-quantization axis leave χ unchanged.

2. **Stability.** Small systematic uncertainties on the measured E_N values produce small changes in χ in bottleneck distance [9].

3. **Structural discrimination.** Two states with identical bipartite entanglements but different multipartite structure can yield different χ. For example, an n-body system with star-graph edge pattern (all edges radiating from one vertex) has a V-shape / tree topology; a system with a cycle structure has $H_1 \neq 0$. These are distinguishable signatures.

5.2 Limitations of the flag-complex construction

The flag complex is blind to genuinely multipartite entanglement that is invisible to pair marginals. We have verified explicitly:

- The n-qubit GHZ state $(|0\cdots0\rangle + |1\cdots1\rangle)/\sqrt{2}$ has $E_N(i{:}j) = 0$ for all pairs. The flag complex has no edges. This is Remark 8 of [1].

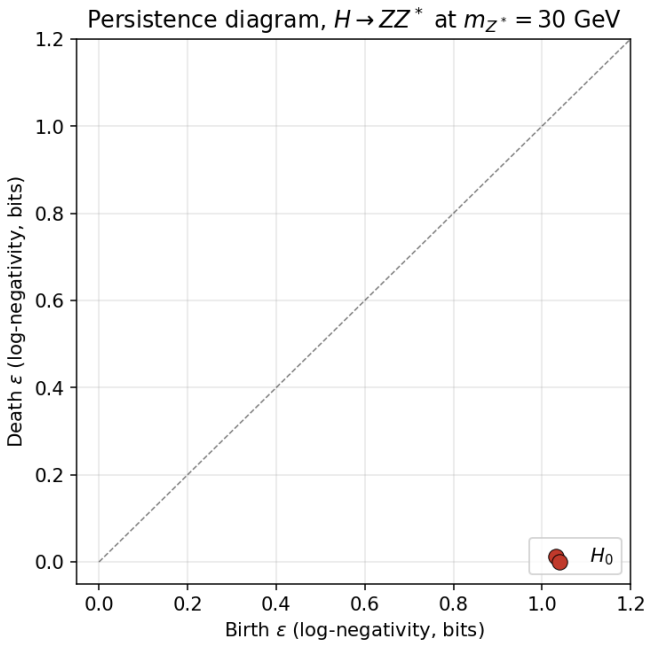

Figure 3: Persistence diagram for $H \to ZZ^*$ at $m_{Z^*} = 30$ GeV. The single H_0 feature has birth $\epsilon \approx 1.04$ and death $\epsilon = 0$ (maximally persistent).

- The 4-qubit linear cluster state $|C_4\rangle$ also has $E_N(i{:}j) = 0$ for all pairs. The flag complex is trivial.
- By contrast, the 3-qubit W state has $E_N = 0.498$ bits for every pair, yielding a complete K_3 flag complex with 2-simplex.

A broader class of stabilizer states with no pair-marginal leakage is mapped to trivial signatures by this construction. Addressing this limitation would require a richer construction — for instance, weighting higher simplices by a genuine multipartite measure that respects face-closure — and remains an open problem. The fact that $H \to ZZ^*$ produces a *non-trivial* flag complex indicates that its entanglement is of the pair-leakage-present type, not GHZ-like.

5.3 Connection to the measurement program

The tripartite density operators of $H \to ZZ^*$ are measurable from angular distributions using the method of [2]. With Run 3 data ($\sim 300\,\text{fb}^{-1}$), the individual E_N values are expected to be determined with statistical precision sufficient to resolve the $L{-}S$ edges from zero at $> 5\sigma$. The $S_1{-}S_2$ edge, being much smaller, is harder; however, the *qualitative* topological signature ("2 large edges, 1 small edge") is robust against the statistical uncertainty even if the small edge cannot be pinned down to better than a factor of two.

Future extensions of this methodology include:

- Application to $t \to Wb$ decays (tripartite L-S_W-S_b system, studied theoretically in [13]).
- Higher-multiplicity systems: four-body final states from $t\bar{t}h$ and $t\bar{t}Z$, where flag complexes on 4+ nodes can exhibit H_1 loops.
- Inclusion of NLO QCD and EW corrections to a_{11}, a_{00}, a_{-1-1} and assessment of their impact on $\chi(K_W)$ [14, 15].

6 Conclusion

We have carried out the first application of the Relational Witnessing Framework's topological construction [1] to a physically-realized multipartite quantum state at the LHC. The methodology is validated on the measured ATLAS $t\bar{t}$ state (log-negativity 0.385 bits), and applied to the tripartite $H \to ZZ^*$ system where it reveals a characteristic L-centric topological signature with persistent V-shaped 1-skeleton and suppressed $S_1 - S_2$ edge. This pattern is a direct topological encoding of angular-momentum-mediated spin entanglement in the Higgs decay. We propose $\chi(K_W)$ as a novel, basis-independent, stable collider observable for characterizing multipartite quantum systems at the highest accessible energy scales.

Code availability. All code and data used in this analysis are available at https://github.com/trevorscott/topological-signatures-entanglement, including the pipeline, the validation tests, and the figure-generation scripts.

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